

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**First Semester of B. Pharm Examination****University Theory Examination March 2018****MA131.1 Basic Mathematics****Date: 31/03/2018, Saturday****Time: 10:00 a.m. to 01:00 p.m.****Maximum Marks: 80****Instructions:**

1. There are three sections in this question paper.
2. SECTION -I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION-I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION-I is 30 minutes. Answers to SECTION-I must be written during the first 30 minutes of the examination.
3. Answers to SECTION-II and SECTION-III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.
6. Use of scientific non-programable calculator is allowed.

SECTION I**Q - 1** Choose the correct answer for the following questions each .**[20]**

1. If E and F are finite sets, then $n(E) + n(F) =$ _____
 (A) $n(E \cup F)$ (B) $n(E \cap F)$ (C) $n(E \cup F) + n(E \cap F)$ (D) $n(E \cup F) - n(E \cap F)$
2. If E and F are two sets, then $E \cap (E \cup F) =$ _____
 (A) E (B) F (C) ϕ (D) $E \cap F$
3. If P and Q are two sets and P' denotes complement of P then $P \cap (P \cup Q)' =$ _____
 (A) P (B) Q (C) ϕ (D) $P \cap Q$
4. Given two non-empty sets E and F , if $n(E) = n_1$ and $n(F) = n_2$ then $n(E \times F) =$ _____
 (A) $n_1 + n_2$ (B) $n_1 \times n_2$ (C) $n_1 - n_2$ (D) $\frac{n_1}{n_2}$
5. If $f(x) = a + bx$, $f(-1) = -5$ and $f(3) = 3$, (where a and b are integers) then a and b are _____
 (A) $a = -3, b = -1$. (B) $a = 2, b = 3$. (C) $a = -3, b = 2$. (D) $a = 2, b = -3$.
6. Let the function $f : \mathcal{R} \rightarrow \mathcal{R}$ be defined by $f(x) = 4x - 3$ for all $x \in \mathcal{R}$ then $f^{-1}(x) =$ _____
 (A) $\frac{x+3}{4}$ (B) $\frac{1}{4x-3}$ (C) $\frac{4}{x} - 3$ (D) $\frac{4}{x+3}$
7. Let the function $f : [2, +\infty) \rightarrow \mathcal{R}$ be defined by $f(x) = x^2 - 4x + 5$ then the range of f is _____
 (A) $\mathcal{R}$ (B) $[1, +\infty)$ (C) $[4, +\infty)$ (D) $[5, +\infty)$
8. Let $f : \mathcal{R} \rightarrow \mathcal{R}$ be defined by $f(x) = \begin{cases} 2x & x > 3, \\ x^2 & 1 < x \leq 3, \\ 3x & x \leq 1 \end{cases}$
 Then $f(-1) + f(2) + f(4) =$ _____

- (A) 9 (B) 5 (C) 14 (D) 10
9. The function $f : \mathcal{R} \rightarrow \mathcal{R}$ defined by $f(x) = \frac{x}{x^2 + 1}$ for all $x \in \mathcal{R}$ is = _____
 (A) onto function. (C) one-one function.
 (B) neither one-one nor onto (D) not onto function.
 function.
10. Sample variance of the data 2,4,5,6,8,17 is 28. Then Sample variance of 4,8,10,12,16,34 is = _____
 (A) 28 (B) 30 (C) 56 (D) 112
11. The average of 7 numbers 7,9,12, x ,5,4,11 is 9. The missing number is = _____
 (A) 13 (B) 14 (C) 15 (D) 8
12. If $\frac{1}{2}$ is a root of the equation $x^2 + kx - \frac{5}{4} = 0$, then the value of k is = _____
 (A) 2 (B) -2 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$
13. A function f is said to be continuous at $x = c$ if
 (A) $\lim_{x \rightarrow c} f(x)$ exist. (C) $f(c)$ exist
 (B) $\lim_{x \rightarrow c} f(x) = f(c)$. (D) all (A),(B) and (C).
14. $\lim_{x \rightarrow 2} (5x - 3) =$ _____
 (A) 10 (B) -3 (C) 7 (D) -5
15. The slope of the curve $y = (x + 1)^3$ at point $x = -2$ is _____
 (A) 3 (B) -1 (C) -3 (D) 0
16. If $y = 2x^3$ then derivative $\frac{dy}{dx} =$ _____
 (A) 6 (B) x^2 (C) $6x^2$ (D) $3x^2$
17. The integration of the function $\int x^5 dx =$ _____
 (A) x^6 (B) $5x^4$ (C) $\frac{x^6}{6} + C$ (D) $x^6 + C$
18. If $P = \begin{bmatrix} 1 & 2 & 3 \\ 4 & -5 & 6 \end{bmatrix}$, and $Q = \begin{bmatrix} 7 & 8 \\ 0 & -9 \end{bmatrix}$ then
 (A) $P + Q$ is defined (C) QP is defined
 (B) PQ is defined (D) $|P|$ is defined
19. If $U = [u_{ij}] = \begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 1 & 0 & 1 \end{bmatrix}$ then minor of $u_{11} =$ _____
 (A) 4 (B) 1 (C) 0 (D) -4
20. If $P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ then $P^{-1} =$ _____

(A) P

(B) $\frac{1}{P}$

(C) P^{-1} is not defined

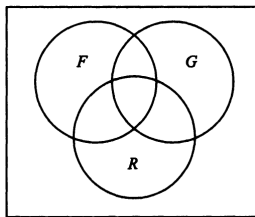
(D) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

SECTION II

Q - 2 Attempt any **Four** of the following.

[20]

- A.** Consider the following data for 120 mathematics students: 65 study French, 20 study French and German, 45 study German, 25 study French and Russian, 42 study Russian, 15 study German and Russian, 8 study all three languages. Let F , G , and R denote the sets of students studying French, German, and Russian, respectively. Fill in the correct number of students in each of the eight regions of the Venn diagram of Fig below.



[05]

- B.** Consider the following relations on the set $P = \{1, 2, 3\}$
 $f = \{(1, 3), (2, 3), (3, 1)\}$, $g = \{(1, 2), (3, 1)\}$ and $h = \{(1, 3), (2, 1), (1, 2), (3, 1)\}$.
Which of the relation is a function from P into P ?

[05]

- C.** Let $E = \{1, 2\}$, $F = \{x, y, z\}$ and $G = \{3, 4\}$ then Find $E \times F \times G$.

[05]

- D.** Evaluate the limits. (a) $\lim_{x \rightarrow -4} \frac{2x + 8}{x^2 + x - 12}$ (b) $\lim_{x \rightarrow -1} \frac{2x^2 + x - 1}{x + 1}$

[05]

- E.** Determine whether the function $f(x) = \frac{x^2 - 4}{x - 2}$ is continuous at $x = 2$.

[05]

- F.** Evaluate the limit $\lim_{x \rightarrow -2} \frac{x^3 + 8}{x + 2}$.

[05]

SECTION III

Q - 3 Attempt any **Four** of the following.

[20]

- A.** Evaluate the integral $\int \frac{t^2 - 2t^4}{t^4} dt$.

[05]

- B.** Evaluate the integral $\int_0^2 \left(1 - \frac{1}{2}x\right) dx$.

[05]

- C.** If $P = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$ and $Q = \begin{bmatrix} c & d \\ d & c \end{bmatrix}$ Show that $PQ = QP$.

[05]

- D.** Find the derivative of $y = 2x^6 + x^{-9}$.

[05]

- E.** If $M = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}$ then compute $|M|$.

[05]

- F.** Find the first and second derivative of $y = 4x^7 - 5x^3 + 2x$.

[05]

Q - 4 Attempt any **Four** of the following. [20]

A. Solve the following system of equations

$$x + 2y + 3z = 6, 2x + 4y + z = 7, 3x + 2y + 9z = 14 \text{ using Cramer's rule.} \quad [05]$$

B. Calculate the variance and standard deviation of following frequency distribution. [05]

Table 1: Frequency distribution				
Lower Limit	4	8	12	16
Upper Limit	7	11	15	19
Frequency	3	6	4	7

C. Find the mean deviation about the median of the following distribution. [05]

Table 2: Distribution of Marks					
Marks Obtained	10	11	12	14	15
Number of Students	2	3	8	3	4

D. Mean and standard deviation of 100 observations were found to be 40 and 10, respectively. If at the time of calculation two observations were wrongly taken as 30 and 70 in place of 3 and 27 respectively, find the correct standard deviation. [05]

E. (a) Solve the system of equations $x + 2y = 4, 3x - y = 5$ using Cramer's rule. [03]

(b) Solve the quadratic equation $x^2 + x - 12 = 0$. [02]

F. The marks obtained by 9 students in a mathematics test are 50, 69, 20, 33, 53, 39, 40, 65 and 59. Compute all quartiles. [05]